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Modern Use of Steganography Using Python

KH4017CMD Introduction to Programming

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Introduction:

Ever since technology emerged, cyber risks and attacks started increasing drastically. According to Sanskriti Jain (2024), an average of 2,200 cyberattacks occur per day: one every 39 seconds. “In 2024, organizations experienced an average of 1,636 cyberattacks per week, representing a 30% year-over-year increase” (Terranova Security,2024). As shown in the previous statistics, cyberattacks are expanding, which results in severe losses in sensitive fields including health, finance, business, etc. For instance, a patient with a specific history with a disease or allergy could be in danger if his/her data got leaked from the hospital, as anyone could use this weak point against him/her. Moreover, assume that a customer's data, including their phone number, has been leaked; any hacker could call the customer and request more information or money, distinguishing as a company or a bank. Therefore, one way to avoid this is by encrypting or hiding the data, ensuring that if a leak occurs, the information remains secure. Consequently, old methods like cryptography and steganography started gaining popularity, but in the modern world. Cryptography is the science behind securing data and communication. It has many techniques like encryption, hashing, digital signatures, and authentication. Encryption is a method under cryptography that's used for exchanging plain text with cipher text, using an encryption and decryption key, to ensure the security of the data and its CIA: confidentiality, integrity, and availability. It has various types (old and new) including Caesar Cipher, Atbash Cipher, RSA, etc. On the other hand, steganography is the science behind hiding a specific file, image, text, audio, or video in another domain: file, image, text, audio, or video. In this report, an application is implemented to take an input from the user, encrypt it, and hide it inside an image, as it aims to ensure the security of the data and its CIA against any threat.

Application:

As mentioned previously, this application aims to secure the user's input by hiding its, ensuring its CIA. It starts by taking the user's input which is the message needed to be hid. This message or value, in the python language, is stored in a variable named "secret_message". The next step is the Encrypting and Decrypting class. First, the class gets all the characters ("strings") that could be used in the user's message which includes all punctuation marks, letters (both uppercase and lowercase), and digits. It then converts these characters to a list named characters: original list. To make our encryption key, a copy is taken from this original list and shuffled. Now, there are 2 lists: "characters" that contains all the original characters and "key" that contains all shuffled characters. The class then has 2 methods: encryption and decryption. The encryption method takes a text (the user's message in this case) as a parameter. Then, a variable called "cipher" is declared with empty string so the encrypted message could be added in. The function will then loop through the unencrypted message, checking if all the letters in the message is included in the characters list or not. If the letter is included, then the function will start shifting each unencrypted character in the characters list with 13 then exchanging it with another character from the key list that has the same shifted index. If the letter was not found in the characters list, the original letter will be added to the cipher variable without any encryption. When the letters end in the text, the loop will stop iterating and the function will return the cipher or encrypted text. The decryption method is the same steps but in reverse. It takes the encrypted text as a parameter and declares a "plain" variable that has a value of an empty string. It then loops thorough every letter in the encrypted message, checking if all the letters in the message is included in the key list or not. If yes, the function will start de-shifting the letter's index by 13 and exchanging it by another letter from the characters, original, list that has the same de-shifted

index. If the letter was not found in the key list, the encrypted letter will be added to the plain variable without any decryption. Finally, it returns the plain text when the loop ends. Now that the class is done, an instance from the class is taken and saved to a variable called “cipher tool” and used for encrypting the user’s message. This encrypted message gets inputted as a parameter in the function that converts text to binary. This function takes the encrypted message, iterates over each character from the message and converts it to its binary ASCII [1] value using the built in function ord(), then to binary(8 bit format). It also uses the join built in python, so all the binary numbers are saved and joined together as one number in one variable. Finally, the function returns the binary value of the encrypted message. Moving on to the “hide_message_in_image” function, this function is where the message gets hid in the image. It takes 3 parameters: the inputted image path, the binary representation of the secret message and the output image path. Using the help of the pillow library, it opens the image, converts it to RGB: red, green, and blue. RGB could be considered as color palette that displays the colors that are utilized on a computer screen. Any color in the visible spectrum can be created by adjusting the amounts of bits of red, green, and blue. Then, the length of the binary message is calculated, putted in a 32-bit integer format, and finally prepending the length with the binary message. Prepending is the process of concatenating(str) the length and the actual binary message, resulting in a 32-bit number that represents the length of the message and the actual message value. Any number of bits could be used; however, large number of bits are better as it provides a wider range of bits (32-bit= 0 to 4,294,967,295). This process is done to ensure that the message’s start and end bits are known. Then, each character in the binary message is separated and converted (either '0' or '1') into an integer, turning them into a list. The function then checks the image size and loops through each pixel. The function will then start looping again in range

of 3(RGB channels); it will check the amount of bits in the message to embed then it will ensure that the least significant bit in each RGB channel is cleared and set to 0 without affecting the other bits then this 0 is exchanged with the message's bit. The pixel is then turned to a tuple to be compatible with the image format. The function will finally show the picture with the hidden message, including its path. Then, the “extract_message_from_image” function will take place. The function takes one parameter which is the output image path. It then opens the image and loops through its columns(x) and rows(y), getting all the pixels. It then loops through the RGB channels, extracting each pixel's LSB and appends it to the extracted_bits list which will eventually hold all the bits of the hidden message. The code extracts the first 32 bits to determine the message's length. It then extracts these bits from the list to get the actual message. Finally, it converts those message bits into a single binary string and returns it. After extracting the binary message from the image, this binary message will be inputted as a parameter for the “binary_to_text” function which converts the binary representation to the actual encrypted message using the help of the ASCII characters. Now that the encrypted message is known, the function “decryption” inside the class will be used to decrypt this message using the steps mentioned previously.

0 011 0000	A 100 0001	N 100 1110	a 110 0001	n 110 1110	. 010 1110
1 011 0001	B 100 0010	O 100 1111	b 110 0010	o 110 1111	, 010 1100
2 011 0010	C 100 0011	P 101 0000	c 110 0011	p 111 0000	! 010 0001
3 011 0011	D 100 0100	Q 101 0001	d 110 0100	q 111 0001	? 011 1111
4 011 0100	E 100 0101	R 101 0010	e 110 0101	r 111 0010	' 010 0111
5 011 0101	F 100 0110	S 101 0011	f 110 0110	s 111 0011	(010 1000
6 011 0110	G 100 0111	T 101 0100	g 110 0111	t 111 0100	) 010 1001
7 011 0111	H 100 1000	U 101 0101	h 110 1000	u 111 0101	- 010 1101
8 011 1000	I 100 1001	V 101 0110	i 110 1001	v 111 0110	" 010 0010
9 011 1001	J 100 1010	W 101 0111	j 110 1010	w 111 0111	space 010 0000
	K 100 1011	X 101 1000	k 110 1011	x 111 1000	
	L 100 1100	Y 101 1001	l 110 1100	y 111 1001	
	M 100 1101	Z 101 1010	m 110 1101	z 111 1010	

Illustration 1: ASCII Table showing the characters and their corresponding binary ASCII values.

The higher the pixel intensity value, the more the brightness of the color.

Color	Color Name	Pixel Intensity Values - RGB Color Space		
		R (Red)	G (Green)	B (Blue)
	Red	255	0	0
	Green	0	255	0
	Blue	0	0	255
	Yellow	255	255	0
	Cyan	0	255	255

Illustration 2: RGB Table showing the colors and their corresponding pixels(bits) values.

Instructions for usage:

1. The image should be bmp.
2. The input and output paths should be different for clarity.
3. The inputted message will be always str.
4. PIL should be installed. (Open a new terminal, write “pip install pillow”)
5. Path format should be like this C:\Users\HALA AHMED\Desktop\year 1\python\original
pic.bmp

Table of unit tests:

In Python, unit testing occurs with the built-in "unittest" module. Unit tests allow you to ensure that each function of the chosen program is performing as expected. To write a unit test, the following stages must be done.

1. Import the "unittest" module.
2. Create a test class that inherits from unittest.TestCase.

3. Write the methods/functions needed for testing within the class, each starting with the word test (so unittest identifies them as tests).
4. Use assertions(assertEqual) to determine whether the code operates as expected.
5. If it behaves as expected: the message "ok" will be shown in the terminal.

```
class TestMessageFunctions(unittest.TestCase): #class that inherits from unittest.TestCase module

    def test_encryption_decryption(self): #the word "test" is added before the function as a pointer to the unittest module
        # Test encryption and decryption results
        cipher_tool1 = Encrypting_and_Decrypting() #instance
        original_message1 = "Hello, World!"
        encrypted_message1 = cipher_tool1.encryption(original_message1)
        decrypted_message1 = cipher_tool1.decryption(encrypted_message1)
        # Assert that after encryption and decryption, the original message is recovered
        self.assertEqual(original_message1, decrypted_message1) #checks that both the input and output is the same/equal

    def test_text_to_binary(self):
        # Test if text is correctly converted to binary
        text1 = "AB"
        expected_binary1 = '0100000101000010' # 'A' = 01000001, 'B' = 01000010
        binary_message1 = text_x_binary_instance1.text_to_binary(text1)
        self.assertEqual(binary_message1, expected_binary1) #checks that both the expected output and the actual output is the same

    def test_binary_to_text(self):
        # Test if binary is correctly converted back to text
        binary_message2 = '0100000101000010'
        expected_text2 = 'AB'
        text2 = text_x_binary_instance1.binary_to_text(binary_message2)
        self.assertEqual(text2, expected_text2) #checks that both the expected output and the actual output is the same

    def test_hide_and_extract_message(self):
        # Create a new image in memory for testing
        original_image = Image.new('RGB', (10, 10), color='white')
        image_path = "hala_image.bmp"
        original_image.save(image_path)

        #Converts the message to its binary representation
        secret_message = "Hidden Message"
        binary_message = text_x_binary_instance1.text_to_binary(secret_message)

        # Hide message in image
```

Illustration 3: Table of unit tests steps from the source code.

```
class TestMessageFunctions(unittest.TestCase): #class that inherits from unittest.TestCase module

    def test_binary_to_text(self):
        # Test if binary is correctly converted back to text
        binary_message2 = '0100000101000010'
        expected_text2 = 'AB'
        text2 = text_x_binary_instance1.binary_to_text(binary_message2)
        self.assertEqual(text2, expected_text2) #checks that both the expected output and the actual output is the same

    def test_hide_and_extract_message(self):
        # Create a new image in memory for testing
        original_image = Image.new('RGB', (10, 10), color='white')
        image_path = "hala_image.bmp"
        original_image.save(image_path)

        #Converts the message to its binary representation
        secret_message = "Hidden Message"
        binary_message = text_x_binary_instance1.text_to_binary(secret_message)

        # Hide message in image
        output_image_path = "output_image.bmp"
        hide_message_in_image(image_path, binary_message, output_image_path)

        # Extract message from image
        extracted_binary_message = extract_message_from_image(output_image_path)
        extracted_message = text_x_binary_instance1.binary_to_text(extracted_binary_message)

        # Assert the extracted message is the same as the original
        self.assertEqual(secret_message, extracted_message)

if __name__ == '__main__':
    unittest.main() #runs all test cases defined in the code, outputting their results
```

Illustration 4: Table of unit tests steps from the source code.

Repository link:

<https://github.com/hala-ahmedd/PYTHON-CW/commits/main/>

Algorithm: in pseudocode

Time complexity is the time taken for an algorithm to output the desired output relatively to the size of its input. The overall time complexity is $O(n + p)$, where: n is the length of the message and p is the number of pixels in the image. The time complexity for the encryption, decryption, and binary conversion functions is $O(n)$ as there's 1 loop. The time complexity for the image-related functions (hiding and extracting the message) is $O(p)$, where p is the number of pixels in the image (height*width).

Overall Application Algorithm

1. The message is taken from the user.
2. The message gets encrypted using The Caesar Cipher Method.
3. The encrypted message is converted to binary.
4. The binary message will be inputted in the image using the LSB approach.
5. The binary message will be extracted from the image using the LSB approach.
6. The binary message gets converted to text.
7. The text gets decrypted.

Encryption Algorithm

1. Import random (for shuffling)
2. Import string (for extracting all characters)
3. Declare variable 'cipher' as empty string to add on
4. For loop to iterate over each letter in the unencrypted text
5. If condition to check if the letter is in the characters list, the following operation will happen
6. Declare variable 'index' = gets the letter's index in the original list which is characters
7. Declare variable 'shifted' = shifts the index by 13 and whenever the list is done, repeat it
8. Add on to cypher text from the shuffled list which is key using the updated index
9. If the letter is NOT in the characters list, leave unsupported characters unchanged
10. The function outputs the encrypted version of the given text

Text to Binary Algorithm

- Text to Binary Algorithm**

 1. Function text to binary that takes a text as a parameter
 2. Initialize an empty string called 'binary'
 3. For loop on each character in 'text'
 4. Convert the character to its ASCII value using ord
 5. Convert the ASCII value to its 8-bit binary representation
 6. Append the binary string to 'binary'
 7. Return the final 'binary' string

Hiding the message

- Hiding the message**

 1. Import pillow
 2. Function that takes 3 parameters (input path, binary message, output path)
 3. Opens the image
 4. Convert the image to RGB
 5. Convert the binary message in a 32-bit format
 6. Convert each bit to an integer
 7. Initialize the number of bits of the message by 0
 8. Loop through all the pixels in the image
 9. Loop through the 3 RGB channels
 10. Check if there are bits to embed
 11. If yes, the LSB will be set to 0 and exchanged with the message's bit
 12. Increment the number of bits
 13. Update the pixels
 14. Save the image in the output path
 15. Show the image
 16. Outputs the message saved output path (new picture with the hidden message)
 17. If not, the function will stop

Example:

encode the bit 1 in pixel value:

 - Original pixel value: 11001010
 - Applying XOR with bit 1, the value becomes 11001011 (LSB is now 0).
 - If message bit is 0, applying XOR with 0, the final pixel value remains 11001010 (which is 202 in decimal).

Example:

encode the bit 1 in the least significant bit of a pixel value:

- Original pixel value: pixel[i] = 255 (in binary: 11111111)
- Applying pixel[i] & 0xFE, the pixel value becomes 11111110 (the LSB is now 0).
- If message bits[bit index] = 1, applying | 1 sets the LSB to 1, so the final pixel value becomes 11111111 (which is 255 in decimal again).

- encode the bit 1 in the least significant bit of a pixel value:
 - Original pixel value: $\text{pixel}[i] = 255$ (in binary: 11111111)
 - Applying $\text{pixel}[i] \& 0xFE$, the pixel value becomes 11111110 (the LSB is now 0).
 - If message bits[bit index] = 1, applying | 1 sets the LSB to 1, so the final pixel value becomes 11111111 (which is 255 in decimal again).

- Original pixel value: $\text{pixel}[i] = 255$ (in binary: 11111111)
- Applying $\text{pixel}[i] \& 0xFE$, the pixel value becomes 11111110 (the LSB is now 0).
- If message bits[bit index] = 1, applying | 1 sets the LSB to 1, so the final pixel value becomes 11111111 (which is 255 in decimal again).

Binary to 32-bit

- Binary to 32-bit**

 1. Take binary message (done as a parameter in the hiding function)
 2. Calculate the length of the binary message
 3. Convert the length (number itself) to a 32-bit binary string
 4. Prepend the length with the actual binary message

Example:

If binary message is "110101001", then the length of the binary message is 9.
The length (9) as a 32-bit binary string=
(00000000000000000000000000001001)

Prepend: concatenate them as strings (length in the beginning)
(00000000000000000000000000001001) + (110101001) =
Binary message = '00000000000000000000000000001001110101001'

Example:
 If binary message is "110101001", then the length of the binary message is 9.
 The length (9) as a 32-bit binary string=
 (000000000000000000000000000000001001)
 Prepend: concatenate them as strings (length in the beginning)
 (000000000000000000000000000000001001) + (110101001) =
 Binary message = '000000000000000000000000000000001001110101001'

If binary message is "110101001", then the length of the binary message is 9.
 The length (9) as a 32-bit binary string=
 (00000000000000000000000000001001)
 Prepend: concatenate them as strings (length in the beginning)
 (00000000000000000000000000001001) + (110101001) =
 Binary message = '00000000000000000000000000001001110101001'

The length (9) as a 32-bit binary string= (000000000000000000000000000001001)

[illegible]

Prepend: concatenate them as strings (length in the beginning)

[illegible]

Binary message = '000000000000000000000000000000001001110101001'

Extracting the message

1. Import pillow (already imported in the hiding function so no need)
2. Function that takes 1 parameter (output path)
3. Gets the image dimensions
4. Initializes an empty list to store extracted bits
5. Loop through all the pixels in the image
6. Loop through the 3 RGB channels
7. Extracts LSB and append it to extracted bits

Example:

- `pixel[i] = 255` (in binary: 11111111).
- `0x01 = 00000001` (a bitmask to isolate the LSB).
- `pixel[i] & 0x01 = 11111111 & 00000001 = 00000001`, so the LSB is **1**.
Now, this extracted bit (1) is appended to the extracted bits list.

32-bit to binary

1. Slice the extracted bits list to get the first 32 elements
2. Convert the length bits (which is a list of bits) into a binary string
3. Converts it to an integer to know the length
4. Extract the message from the list of bits, based on the length information stored in the first 32 bits (length+32)
5. Convert each element of message bits to a string
6. Concatenate them into a single binary string
7. Return binary message

Decryption Algorithm

11. Import random (for shuffling)
12. Import string (for extracting all characters)
13. Declare variable as 'plain' empty string to add on
14. For loop to iterate over each letter in the encrypted text
15. If condition to check if the letter is in the key list, the following operation will happen
16. Declare variable 'index' = gets the letter's index in the key list
17. Declare variable 'shifted' = de- shifts the index by 13 and whenever the list is done, repeat it
18. Add on to plain text from the original list which is characters using the updated index
19. If the letter is NOT in the key list, leave unsupported characters unchanged
20. The function outputs the decrypted version of the given text

Binary to Text Algorithm

1. Function binary_to_text that takes binary as a parameter
2. Initialize an empty string called 'message'
3. For loop: `i` from 0 to length of 'binary' with a step of 8:
4. Extract a substring of 8 bits from 'binary'
5. If the substring length is 8:
6. Convert the 8-bit binary substring to an integer
7. Convert the integer to its corresponding character
8. Append the character to 'message'
9. Return the final 'message' string

Source code:

```
python-coursework.py X
C: > Users > HALA AHMED > Desktop > year 1 > python > uni > PYTHON-CW > python-coursework.py > ...
1  ...
2  GOAL: This application aims to hide a secret message ,given by the user, in a picture.
3  ALGORITHM/STEPS:
4  1. The message is taken from the user.
5  2. The message gets encrypted using The Caesar Cipher Method.
6  3. The encrypted message is converted to binary.
7  4. The binary message will be inputted in the image using the LSB approach.
8  5. The binary message will be extracted from the image using the LSB approach.
9  6. The binary message gets converted to text.
10 7. The text gets decrypted.
118. The functions get tested.
12...
13#Importing the needed libraries for Caesar Cipher Encryption and Decryption implementation
14import random #a module for generating random numbers and performing random operations including shuffling
15import string #a module that contains all the characters, useful for working with functions/ with strings needed
16#Importing the image processing library needed to manipulate the picture
17from PIL import Image #PIL= pillow (should be installed in a new terminal with the "pip install Pillow")
18import unittest # a module to verify that individual functions in the program work as expected
19
20# Caesar Cipher Encryption and Decryption Class:
21class Encrypting_and_Decrypting:
22    #Class's shared attributes
23    characters = string.punctuation + string.ascii_letters + string.digits #gets all the characters
24    characters = list(characters) #converts them to a list
25    characters.append(" ") # add space as a character to the list
26    key = characters.copy() #takes a copy of the character's list and calls it key
27    random.shuffle(key) # shuffles the key
28
29    # Encryption Method
30    def encryption(self, text): #self: serving as a pointer , "text"= hidden message
31        cipher = "" #empty string to add on the encrypted characters
32        for letter in text: #iterates over each letter in the unencrypted text
33            if letter in self.characters: #check if the letter is in the characters list, the following operation will happen
34                index = self.characters.index(letter) #gets the letter's index in the original list which is characters
35                shifted = (index + 13) % len(self.characters) # shifts the index by 13 & whenever the list is done, repeats it
36                cipher += self.key[shifted] #add on to the cipher variable from the shuffled list which is key using the updated index
37            else: #if the letter is NOT in the characters list, the following operation will happen
```

```
python-coursework.py X
C: > Users > HALA AHMED > Desktop > year 1 > python > uni > PYTHON-CW > python-coursework.py > ...
21 class Encrypting_and_Decrypting:
30     def encryption(self, text): #self: serving as a pointer , "text"= hidden message
37         else: #if the letter is NOT in the characters list, the following operation will happen
38             cipher += letter # leave unsupported characters unchanged
39         return cipher #the function outputs the encrypted version of the given text
40
41     # Decryption Method
42     def decryption(self, cipher):
43         plain = "" #empty string to add on the decrypted characters
44         for letter in cipher: #iterates over each letter in the encrypted text
45             if letter in self.key: #check if the letter is in the key list, the following operation will happen
46                 index = self.key.index(letter) #gets the letter's index in the copied, shuffled list: key
47                 unshifted = (index - 13) % len(self.characters) #de-shifts the index by 13 & whenever the list is done, repeat it
48                 plain += self.characters[unshifted] #add on to plain text from the original list which is characters using the updated index
49             else:
50                 plain += letter # leave unsupported characters unchanged
51         return plain #the function outputs the decrypted version of the given text
52
53 #Text and Binary conversions class:
54 class Conversion:
55     # Convert text to binary method
56     def text_to_binary(self, text):
57         # convert each character in the text to its 8-bit binary representation using its ascii and joins them into one string
58         binary = "".join(format(ord(i), '08b') for i in text)
59         return binary
60
61     # Convert binary to text method
62     def binary_to_text(self, binary):
63         message = "" #empty string to add on
64         for i in range(0, len(binary), 8): #increments by 8 each time
65             byte = binary[i:i + 8] # Extracts 8 bits (1 byte) each iteration
66             if len(byte) == 8: #ensures that the bits are equal to 8
67                 message += chr(int(byte, 2)) #converts the byte binary values to int values then changes them as characters
68         return message # outputs the final decoded message
69
70 # Hide Message in Image Function
71 def hide_message_in_image(image_path, binary_message, output_image_path):
```

```

python-coursework.py X
C: > Users > HALA AHMED > Desktop > year 1 > python > uni > PYTHON-CW > python-coursework.py > ...
69 # Hide Message in Image Function
70 def hide_message_in_image(image_path, binary_message, output_image_path):
71     image = Image.open(image_path) #opens the image
72     image = image.convert("RGB") #converts the image to RGB (Red,Green,Blue) to have bits based on the shades of the pictures
73
74     # Include the length of the binary message as a 32-bit integer (from 0 to 4,294,967,295) to know the message's limits
75     message_length = len(binary_message) #calculates how long(length) the binary message is (in bits).
76     length_binary = format(message_length, '032b') # puts the length of the message in 32-bit length format
77     binary_message= length_binary + binary_message # Prepend length(adding it to the beginning of a string)
78
79     message_bits = list(map(int, binary_message)) #converts each character (either '0' or '1') in the binary message into an integer and turns them into a list
80     width, height = image.size # gets the width and height of the image in pixels, so we know how many pixels are available for embedding the message
81     bit_index = 0 #intilaizing the number of bits of the message
82     for y in range(height): #iterate over every pixel in the image (width by height).
83         for x in range(width): #iterate over every pixel in the image (width by height).
84             pixel = list(image.getpixel((x, y))) #get the pixel at position (x, y) as a list of tuples of (R, G, B) values
85             for i in range(3): # 3 RGB channels
86                 if bit_index < len(message_bits): #checks if we still have bits left in the message to embed
87                     pixel[i] = (pixel[i] & 0xFE) | message_bits[bit_index] #this operation ensures that the LSB is cleared (set to 0) without affecting the other bits and
88                     bit_index += 1 #increments the index to move to the next bit in the message.
89             image.putpixel((x, y), tuple(pixel)) # the pixel is updated using image.putpixel((x, y), tuple(pixel)) where tuple(pixel) converts the list back to a tuple t
90             if bit_index >= len(message_bits): #if all bits of the message have been embedded, the loop breaks early to avoid unnecessary pixel processing.(for the neste
91                 break
92             if bit_index >= len(message_bits): #if all bits of the message have been embedded, the loop breaks early to avoid unnecessary pixel processing.(for the original
93                 break
94     image.save(output_image_path) #saves the image in the desired path
95     image.show() #shows the picture to the user
96     print(f"Message hidden in image: {output_image_path}")
97
98 # Extract Message from Image Function
99 def extract_message_from_image(image_path):
100     image = Image.open(image_path) # opens the image
101     width, height = image.size # gets the image dimensions
102     extracted_bits = [] # initializes an empty list to store extracted bits

```

```

python-coursework.py X
C: > Users > HALA AHMED > Desktop > year 1 > python > uni > PYTHON-CW > python-coursework.py > ...
99 def extract_message_from_image(image_path):
104     for x in range(width):#iterate over every pixel in the image (width by height)
105         pixel = list(image.getpixel((x, y)))#get the pixel at position (x, y) as a list of tuples of (R, G, B) values
106         for i in range(3): # 3 RGB channels
107             extracted_bits.append(pixel[i] & 0x01) # extracts LSB and append it to extracted_bits
108
109     # Extract the first 32 bits for the message length
110     length_bits = extracted_bits[:32] # slicing the list to get the first 32 elements
111     message_length = int("".join(map(str, length_bits)), 2) #converts the length_bits (which is a list of bits) into a binary string, then converts it to an integer.
112
113     # Extract the actual message bits
114     message_bits = extracted_bits[32:32 + message_length] #extracts the message from the list of bits, based on the length information stored in the first 32 bits.
115     binary_message = "".join(map(str, message_bits)) #converts each element of message_bits to a string and concatenate them into a single binary string
116     return binary_message
117
118 # User Code
119 if __name__ == "__main__": #main function: all functions start/get provoked after it (could be removed but better for clarity and organization)
120     # Step 1: User Input
121     secret_message = input("Enter your secret message: ") #the input is taken as a string
122
123     # Step 2: Encrypt the Message
124     cipher_tool = Encrypting_and_Decrypting() #instance from the class
125     encrypted_message = cipher_tool.encryption(secret_message) #encrypting the message using the function in the class's instance
126     print(f"Encrypted Message: {encrypted_message}")
127
128     # Step 3: Convert Encrypted Message to Binary
129     text_x_binary_instance1=Conversion() #instance from the class
130     binary_message = text_x_binary_instance1.text_to_binary(encrypted_message) #converting the message using the function in the class's instance
131     print(f"Binary Representation: {binary_message}")
132
133     # Step 4: Hide Binary Message in Image
134     input_image_path = input("Enter input image path please: ") # user's original image path
135     output_image_path = input("Enter output image path please: ") # user's output image path
136     hide_message_in_image(input_image_path, binary_message, output_image_path) #calling/ provoking the function
137
138     # Step 5: Extract Binary Message from Image
139     extracted_binary_message = extract_message_from_image(output_image_path) #calling/ provoking the function while saving its return value in a variable

```

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python-coursework.py X
C:\Users\HALA AHMED\Desktop\year 1\python> uni PYTHON-CW > python-coursework.py > ...

138 # Step 5: Extract Binary Message from Image
139 extracted_binary_message = extract_message_from_image(output_image_path) #calling/ provoking the function while saving its return value in a variable
140 print(f"Extracted Binary Message: {extracted_binary_message}")
141
142 # Step 6: Convert Binary to Text
143 extracted_encrypted_message = text_x_binary_instance1.binary_to_text(extracted_binary_message) #converting the message using the function in the class's instance
144 print(f"Extracted Encrypted Message: {extracted_encrypted_message}")
145
146 # Step 7: Decrypt the Extracted Message
147 decrypted_message = cipher_tool.decryption(extracted_encrypted_message) #decrypting the message using the function in the class's instance
148 print(f"Decrypted Message: {decrypted_message}")
149
150 class TestMessageFunctions(unittest.TestCase): #class that inherits from unittest.TestCase module
151
152 def test_encryption_decryption(self): #the word "test" is added before the function as a pointer to the unittest module
153     # Test encryption and decryption results
154     cipher_tool1 = Encrypting_and_Decrypting() #instance
155     original_message1 = "Hello, World!"
156     encrypted_message1 = cipher_tool1.encryption(original_message1)
157     decrypted_message1 = cipher_tool1.decryption(encrypted_message1)
158     # Assert that after encryption and decryption, the original message is recovered
159     self.assertEqual(original_message1, decrypted_message1) #checks that both the input and output is the same/equal
160
161 def test_text_to_binary(self):
162     # Test if text is correctly converted to binary
163     text1 = "AB"
164     expected_binary1 = '0100000101000010' # 'A' = 01000001, 'B' = 01000010
165     binary_message1 = text_x_binary_instance1.text_to_binary(text1)
166     self.assertEqual(binary_message1, expected_binary1) #checks that both the expected output and the actual output is the same
167
168 def test_binary_to_text(self):
169     # Test if binary is correctly converted back to text
170     binary_message2 = '0100000101000010'
171     expected_text2 = 'AB'
172     text2 = text_x_binary_instance1.binary_to_text(binary_message2)
173     self.assertEqual(text2, expected_text2) #checks that both the expected output and the actual output is the same
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